

JUSTIN KIM

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Education

University of Wisconsin–Madison | GPA: 3.76

B.S. in Computer Science & Data Science

Expected Graduation: Dec 2026

Madison, WI

Georgia Institute of Technology

M.S. in Computer Science, Machine Learning Specialization

Incoming Spring 2027

Online, Part-Time

Experience

Responsible AI for Learning (TRAIL) Lab | UW–Madison

Sep 2026 – Present

Undergraduate Research Assistant

Madison, WI

- Developing an MCP-based multi-agent Q&A system with cross-tool retrieval and access controls.
- Red-teaming the agent against prompt injection, data leakage, and unauthorized tool use before deployment.

Capital One

Jun 2026 – Aug 2026

Software Engineer Intern

Chicago, IL

- Built an AI tool generating card fulfillment workflows from plain-English specs, cutting 2+ days to under an hour.
- Designed a GraphRAG pipeline grounding LLM output in approved actions and enforcing config schemas, reaching 100% structural validity and 88% required-field completeness across 70 configs.
- Implemented real-time graph validation in a React/Monaco editor, detecting cycles, unreachable states, and invalid references; adopted by 20+ intent engineers.

MadLease

Apr 2025 – Present

Co-Founder & Founding Engineer

Madison, WI

- Founded and grew a UW–Madison sublease marketplace to 250+ accounts, 100+ listings, and 5+ building partners.
- Replaced recency-based search with PostgreSQL ranking across rent, campus proximity, and 8+ user filters.
- Moved image uploads to browser-direct S3 via presigned URLs, with retry and rollback across 1,000+ uploads.

Integral

May 2025 – Aug 2025

Full Stack Software Engineer Intern

Madison, WI

- Replaced hard-coded crash-intake logic with per-workflow rules, letting providers define claim requirements.
- Engineered an Angular/Tesseract.js OCR module that re-read VIN, plate, and odometer in a background worker until each passed format and lookup validation, reaching 90%+ accuracy and company-wide reuse.

Denari

Feb 2025 – May 2025

Software Engineer, Capstone Project

Madison, WI

- Built a citation-grounded tax RAG system indexing 50,000+ documents using OpenAI embeddings and pgvector.
- Achieved 61% accuracy on 2,200+ CPA exam questions, outperforming GPT-4o-mini by 5 percentage points.

Astro AI Trading

Jan 2024 – Jan 2025

Machine Learning Engineer

Remote

- Led 3 engineers developing forecasting models using technical, macroeconomic, and BERT-derived sentiment features, reducing held-out RMSE by 16% vs. a price-only baseline.
- Fine-tuned a BERT 3-class news-sentiment classifier (positive/negative/neutral), reaching 77% test accuracy.

Projects

QuizClash | *Devpost* | *Source Code*

Node.js | **React** | **WebSocket** | **AWS**

- Built a real-time multiplayer quiz app converting PDFs and YouTube videos into Kahoot-style games via WebSockets.
- Automated quiz generation with Whisper, T5, RoBERTa, and Sense2Vec, achieving 75% manually evaluated relevance.

Technical Skills

Languages: Python, Java, JavaScript/TypeScript, Go, C, SQL, HTML/CSS

Libraries/Frameworks: React, Spring Boot, Node.js/Express, Flask, PyTorch, Angular, Scikit-Learn

Tools: Git, Docker, AWS (S3, EC2, Lambda, CloudFront, IAM, CDK), PostgreSQL, TimescaleDB/pgvector, Firebase